DILIP P. KURDIYA

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SUMMARY

Gameplay Programmer with 10 years of game development experience, including 7+ years specializing in Unreal Engine development. Experienced in designing scalable gameplay systems, gameplay architecture, character frameworks, RTS systems, custom vehicle systems, and performance optimization using Unreal Engine C++ and Blueprint. Delivered gameplay solutions across PC, mobile, and web platforms through freelance and independent projects, with a strong focus on clean architecture, scalable system design, maintainability, and high-quality gameplay experiences.

CORE EXPERTISE

Gameplay Programming • Unreal Engine C++ • Gameplay Architecture • Character Framework • RTS Systems • World State Manager • Animation Systems • Performance Optimization

TECHNICAL SKILLS

Programming Languages	C++, Blueprint, C#
Game Engines	Unreal Engine 5, Unreal Engine 4, Unity
Gameplay Programming	Gameplay Framework, Gameplay Architecture, Character Framework, Animation Systems, Input Systems, Camera Systems, Save & Load Systems, UI Programming, Puzzle Systems, RTS Systems, Custom Vehicle Systems, World State Manager
AI	Behavior Trees, EQS, AI Controllers, Navigation System, Perception System
Rendering & Environment	Niagara, Material Systems, Material Parameter Collections (MPC), Material Instances, Lighting and Level Streaming, Level Design
Multiplayer	Replication, RPCs, Client-Server Architecture
Performance & Optimization	CPU Optimization, Memory Optimization, Unreal Insights, Profiling, Gameplay Performance Optimization
Tools	Visual Studio, Git, Perforce, Blender, Autodesk Maya, Adobe Photoshop

EXPERIENCE

Gameplay Programmer

Self-Employed | 2016 – Present

- Delivered gameplay programming solutions using **Unreal Engine C++, Blueprint, Unity, and C#** across PC, mobile, and web platforms.
- Designed and developed modular gameplay systems with a strong emphasis on scalability, maintainability, clean architecture, and performance optimization.
- Collaborated with freelance clients and multidisciplinary development teams to implement gameplay features, debug technical issues, and deliver production-ready solutions.
- **Delivered gameplay solutions across multiple genres**, including real-time strategy, adventure, puzzle, simulation, shooter, racing, tower defense, and casual games.
- Managed the gameplay development lifecycle from feature planning and implementation through testing, optimization, and client delivery.

SELECTED PROJECTS & FREELANCE WORK

Aatma (Working Title)

Independent Unreal Engine Project

Third-Person Adventure | Puzzle | Exploration | PC | Unreal Engine 5.7 | C++

Developed a third-person adventure game focused on exploration, environmental puzzles, and immersive world building. Contributed gameplay programming, gameplay architecture, dynamic world systems, rendering-related systems, environment systems, and performance optimization while collaborating within a two-person development team.

Key Contributions

- Designed and implemented modular and reusable gameplay systems using Unreal Engine C++ and Blueprint, with a strong emphasis on clean architecture, scalability, maintainability, and performance.
- Developed a **modular player character framework** featuring advanced locomotion, realistic movement, rotation mechanics, input handling, and seamless integration with gameplay systems and animation.
- Designed and implemented a comprehensive **Animation Blueprint system**, including animation state machines, blend spaces, animation transitions, and detailed character animations to deliver responsive and realistic player movement.
- Engineered a centralized **World State Manager** that dynamically synchronized environmental lighting, fog, material parameters, and gameplay-driven world transitions, providing a scalable foundation for environmental state management.
- Implemented a **Save and Load System** to persist player progression and world-state information across gameplay sessions.
- Designed and implemented core gameplay mechanics supporting exploration and puzzle-solving.
- Implemented **Level Streaming** to efficiently load and unload environments, improving memory usage and runtime performance.
- Developed reusable material systems using **Material Instances** and **Material Parameter Collections (MPCs)** to support dynamic environmental transitions driven by the World State Manager.
- Developed optimized **Niagara systems** to enhance environmental atmosphere and overall gameplay presentation.
- Designed and built complete gameplay environments, including **level layout, landscape composition, lighting, environmental storytelling, and asset placement** to support exploration, puzzle mechanics, and player progression.
- Profiled, debugged, and optimized gameplay systems to improve runtime performance, stability, and maintainability throughout development.
- Collaborated within a two-person development team throughout the complete game development lifecycle, from feature planning and implementation to testing and iteration.

RTS Military Game (Freelance Project)

Real-Time Strategy / Third-Person Action | PC | Unreal Engine 4/5 | C++

Developed gameplay systems for a military real-time strategy game featuring seamless transitions between RTS and direct character control. Contributed gameplay programming, character systems, custom vehicle systems, RTS mechanics, and gameplay architecture while collaborating within a multidisciplinary team of up to 16 developers.

Key Contributions

- Designed and implemented a **hybrid RTS and Third-Person gameplay framework**, enabling players to seamlessly switch between strategic RTS gameplay and direct control of any soldier or vehicle.
- Developed a custom **player character framework** featuring first-person and third-person control, military shooting mechanics, advanced locomotion, and detailed Animation Blueprint systems.
- Designed and implemented a **custom military vehicle system** from scratch using the Pawn framework, including realistic physics-based movement, suspension, torque, force-driven locomotion, weapon systems, and support for both wheeled and tracked vehicles.
- Developed a **vehicle formation system** enabling multiple vehicles to maintain dynamic formations while following a designated leader vehicle during RTS movement commands.
- Engineered a **custom RTS unit command system** supporting unit selection, deselection, highlighting, navigation-based movement, squad splitting and merging, individual unit commands, and formation management.
- Engineered a complete **RTS camera system** supporting strategic navigation, unit management, and seamless transitions between RTS and character control.
- Implemented a unified RTS command interface enabling control of both infantry units and military vehicles.
- Integrated Unreal Engine's Navigation System to support intelligent unit movement and RTS command execution.
- Profiled, debugged, optimized, and maintained gameplay systems to improve runtime performance and stability.
- Collaborated within a multidisciplinary team of up to 16 developers, contributing to gameplay feature implementation and system development.

Additional Freelance Projects

Unity & Unreal Engine | PC, Mobile & Web | C#, C++, Blueprint

Developed gameplay programming solutions across **11+ freelance and personal projects** for PC, mobile, and web platforms, spanning genres including shooters, simulators, strategy, puzzle, tower defence, racing, and casual games.

Key Contributions

- Developed gameplay systems, player controllers, game mechanics, UI, save systems, camera systems, input handling, AI, physics, and animation.
- Implemented mobile platform features including touch controls, advertisements, in-app purchases, Google Play Services, and leaderboards.
- Customized and extended Unity and Unreal Engine projects to meet client-specific gameplay and technical requirements.
- Collaborated directly with freelance clients throughout feature implementation, debugging, optimization, and project delivery.

EDUCATION

Bachelor of Science in Information Technology
University of Mumbai
Graduated: 2016

ADDITIONAL INFORMATION

- Available for Remote Opportunities Worldwide
- Open to Full-Time, Contract, and Freelance Roles
- Passionate about Gameplay Programming, System Architecture, and Performance Optimization.